Health and Fitness Mobile Application – UX Design

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Chapter 1 - Introduction

Overview

The aim and objective of the mobile health and fitness application project is to improve the user engagement and motivational experience of the application through the use of User Experience (UX) design. As a result, a mobile application provides a complete solution through which a user can make significant progress in effectively meeting all of the health and fitness goals. By following UX principles, it is possible to create an approach that revolves around the user and through which the application can make significant progress in achieving a variety of functional goals (Hussain, 2017). Through this, a person can accurately track all his daily activities and plan his food intake. On the other hand, by following this same system, a person can monitor the progress being made in his healthcare by following an easy-to-use interface. All these features give a person the opportunity to use an app that not only helps them in completing various healthcare journeys but also gives them the motivation they need to stick with the app and create a sustainable healthy lifestyle (Dove, 2017). It plays a role in the case. Therefore, by properly using all the principles related to UX design, the various parties related to the project can learn about the various existing common issues and these help them in improving the various experiences related to the user.

Hence this project aims to develop a fully functional mobile application which will be specially targeted at the health and fitness conscious group. It will be ensured that the application follows various common standards. Through this, in creating the application, the necessary wireframes and prototypes will be built for each step in the development that will be used. All these factors will help in carrying out design improvement activities in a cyclical manner for each screen and they will play an important role in successfully executing an application by conducting necessary tests (Moore, 2017). However, there will be various milestones associated with this project and these will be evaluated based on usability heuristics. Here the complete IT application will be built and tests will be performed on it. All these factors will be considered as a critical step in making the project development possible and will play an important role in improving the mobile health and fitness application experience and its functionality.

Current Situation

Healthcare and wellness activities can be analysed in different ways in terms of the applications that are related to current health and fitness. But in this case one of the characteristic exists to provide the users who are related to the use of the application with proper motivation and other necessary help so that they can perform long-term engagement with these applications. Therefore, a common problem exists for all these applications and that is to create a user-friendly experience for the customer and to take steps to solve it by identifying the frustrations and bridge engagement issues that the user creates (Mullins, 2015). It is possible to use UX design-related issues to solve all the gaps in the field of research and through this, steps have to be taken to create an engagement platform if it helps in improving. Because users who want to improve their health and fitness, they face a variety of problems, and crucially, it creates problems in setting goals and tracking goals. For example, due to all these reasons, the plans that a user has regarding food consumption can be of different types in this regard. On the other hand, they may experience problems in monitoring the improvements that users make in performing their daily activities. To eliminate all these issues a user needs to carry out designs keeping in mind that if these activities are not carried out, then a user can completely refrain from using various applications. All these factors can completely hinder a user from achieving their goals and the user can stop using the application altogether (Hoehle, 2016). On the other hand, all the locations and applications that exist in the existing market may face problems in properly integrating the features that are related to different applications. This can lead to various barriers to addressing all of a user's issues through a single platform, and these can significantly impact the different routes that students take to wellness.

Problem Statement

The world of health and fitness apps that exists today faces a variety of obstacles in properly incorporating customer engagement features. All of these factors can fail to create a user-friendly experience for the customer, and at the same time, it can create various forms of frustration and disengagement among users. All these issues can create various challenges for the user and thereby a user may face various problems in adopting different meal plans setting his goals or monitoring his daily progress (Habermann, 2016). All of these factors can affect a person's motivation level and hinder their wellness journey. Because a person may face different types of health and wellness problems at the same time and for, they need to

use different types of applications. All these factors further complicate their health and fitness issues and contribute to inefficiencies and inefficiencies in these activities (Silvennoinen, 2014).

Possible Solutions

To overcome the various problems mentioned above, efforts have been taken to design a mobile application that will be able to integrate various types of health and fitness related features. Through this mobile app, a user can create routines to accurately understand various things including determining his various symptoms tracking his daily activities or planning meals. On the other hand, the mobile application will have various gamification or motivational features and these will increase customer engagement with the application and encourage the customer to use the application for a long time by taking a sustainable step. Through this, the customer will be provided with various batch and developmental rewards and various social challenges and personal recommendations will play an important role in keeping a customer engaged with the application for a long time (Valentim, 2017). On the other hand, this application will take steps to provide personal and customer data for the customer and through this a person can plan different types of food based on his physical condition or take proper control of various things including recommendations as needed. Effective progress can be made. Finally, a phased process will be followed to include mobile applications that feature various smart devices and wearable devices at the same time. All these factors will significantly assist the user in taking steps to maintain better health by improving the function of the application (Gray, 2018).

Aim

This project aims to develop a mobile help and fitness application that addresses the challenges faced by the user by highlighting UX design issues and takes the necessary roles to increase engagement with the application.

Objectives

- To create a visually appealing and user-friendly mobile application related to health and fitness that helps users of different levels of background to easily use the application.

- To use a variety of functions within the mobile application and taking steps to coordinate user health and fitness activities through it
- To make analysis on a variety of social theories and gamification in mobile health and fitness apps, motivating users to stay with the app for a long time

Report Roadmap

A roadmap has been used to phase out the various parts of this project. As part of this, a role is initially created and through this role is accepted to provide various information related to the application. Next, a literature review was used to collect the existing data and later, by using research methods, the research process was analysed. The next part of the project discusses methods for conducting this research. In the next part, the system is designed and is evaluated based on Nielsen's 10 heuristics and WCAG (Yang, 2018). Finally, conclusion and recommendations have been provided by integrating various parts of the research. All these issues have played an important role in coordinating research and various work processes.

Chapter 2 - Literature Review

Introduction

The literature review for this part of the project report has been carried out and multi-pronged steps have been taken to build upon the existing research on the subject. As a result, this literature review reviews the various themes that exist above on almonds and these contribute significantly to the knowledge and understanding of the various topics.

Importance of Health and Fitness-Related Mobile Apps

According to Levy (2015), Mobile health and fitness-related applications have become increasingly popular in recent times, and they significantly impact the customer experience by performing various activities such as monitoring health-related issues and managing wellness. These mobile applications also help in providing a variety of ways for the general public to get various personal life feedback and guidance through which a customer can determine their daily physical activity, food intake, etc. However, many users face various problems in using health and fitness-related applications properly. All of these factors significantly affect the customer experience and fail to provide the necessary motivation to engage with the application. All these issues are considered as a kind of hindrance in mobile application management and they create important challenges for the user in terms of nuts and bolts design issues. But differently, if the mobile applications take effective measures to meet the needs of the customers, they can significantly help the organization in increasing the relationship of the mobile application with the customer keeping them engaged with the organization and increasing overall efficiency. can This report reviews various issues related to UX (user interface) design related to health and fitness applications and how the user experience can be improved by creating a proper interface that engages the user By taking steps to relate them to applications, significant progress can be made in knowing and understanding those subjects (Bunian, 2021). As a result of this, by getting detailed information about the UX (user interface) of mobile applications from all the existing research, various types of mistakes can be identified. These will help the organization in achieving a healthy target in the long run and by using the factors derived from this it will be possible to create various user-centric solutions.

User Experience (UX), and Usability in Mobile App

From the perspective of Neil (2014), UX (user interface) issues play an important role in using mobile applications and they contribute effectively in designing and building various types of mobile applications. By using all these factors, it is possible to manage various important developments including achieving user satisfaction and increasing their engagement and achieving their satisfaction. This requires a comprehensive approach to improve the UX (user interface) and include all user perfectionism and emotional and behavioural issues in a complete journey map. These topics will include creating an information architecture through various designs and multi-faceted interactions through which different types of content can be displayed to the user to formulate the right strategy (Weichbroth, 2020). The main objective of all these activities is to create a fine user experience, which can significantly help the user in improving his application usage activities. By using all these topics in advance, a user can use his application correctly and efficiently and can use it to learn and remember different types of topics efficiently. At the same time, by using UX (user interface) issues, a user can avoid any kind of mistakes in his fitness and wellness issues and these can help in increasing their satisfaction.

Canziba (2018) stated that, by using UX (user interface) concepts, the overall satisfaction of mobile applications can be increased and they can help the organization in bringing about a positive outcome. Therefore, the organization must first identify all the needs and preferences and needs of the customer by using design features related to UX (user interface). As part of this work, it is necessary to gain an accurate understanding of the behaviours of mobile application users and to design applications accordingly so that they can meet the user's mental model and needs (Unger, 2023). All of these factors serve as a foundation for the design and help create a more immersive user experience. So designing a mobile application requires taking various steps and among them making efforts to know and understand user information by doing flaming or photo typing or performing various types of experiments (Vermeeren, 2016). By using all these factors, the children who are using the mobile application can be accurately identified and they can help increase customer satisfaction by following the optimized navigation process.

According to Kushendriawan (2021), to design a mobile application, the design must use the features related to UX (user interface) design properly. As part of this, the user is required to use a simple clean and safe design method, through which the designer can influence the

results of the system. Through this, the designer can work to control the cognitive lead that he has and can take measures to display the information in a clear and aligned manner. By using these familiar design patterns and conventions, the user can proceed to influence the interaction of the various screens in his work. All these factors create a kind of user experience and provide the user with the opportunity to view and understand these applications without any problem (Lestari, 2014). Through this, a user can perform the activities related to using the application properly and have all the functions and other features of the application. By using them, a mobile application can advance the UX (user interface) aspect of the user experience and help in integrating them perfectly with the user application. Through this, a user can increase the retention rate of his application and this can significantly help the application in achieving customer satisfaction.

Existing UX Guidelines for Mobile App Design

Choi (2017) mentioned that There are various types of information for creating applications by using various design techniques related to UX. By using all this information a designer can be able to create a complete application correctly and this can make important progress in increasing the standard of the application by combining it with a user-friendly interface (Hartson, 2018). These generally need to be developed at different times in coordination with existing industry standards and practical experience. Designing mobile applications dealing with UX issues may require taking on some specialized roles.

According to Kalimullah (2017), To build a mobile application by using UX design features, the user must be sure to have a clear navigation menu and other options. In front of all these things a user can demonstrate the correct clean and simple way and these can help in increasing their acceptability. As part of these tasks, users need to use different patterns and these include creating tab bars and adding various navigation or bottom navigation features to different levels of the application. Through this, a user can move between different parts of the application without any problem and it helps them to interact with the application (Yu, 2016). On the other hand, the content that is displayed by using the UX feature inside the mobile application, should be rich in the correct text, shape and design. By following the various types of mobile devices, steps have to be taken to ensure that all these applications work even in the case of changing screen sizes. Part of this process involves sizing the front and ensuring that the light has the right amount of brightness and contrast. On the other hand, all the blank spaces in the text have to be followed properly and they significantly help a

reader to read the information and learn different parts from it (Kascak, 2014). Especially for those users who have various limitations and are visually impaired, these application features help them effectively.

Moran (2018) reviews that, In order to design as many different parts of the application as possible, it is necessary to implement elements related to different colours and shapes and different features of the UX (user interface) design features following the activities of different parts. All these things help in coordinating a proper information process and following various types of information step by step helps in gaining details about the interface of the application and other navigational matters. Through this, it is possible to highlight various important information to the user and it helps the user to navigate and navigate the various interfaces of the application correctly. Again a user can use different platforms while using his mobile application and they can interact with the application using an Android or iOS platform (Yang, 2016). Through this diversification of the application is done and it is possible to take steps to integrate the application with different platforms. Designing mobile applications must take steps that can help provide information to the user that is specific to the platform. By using all these factors a user can confidently and efficiently perform operations between different parts of the application. Through this, a user can increase his engagement due to the UX (user interface) designs of the mobile application and these help the user to engage with the platform for a long time. But in this case, the user has to keep in mind that the various parts of these applications are optimized and they can help in improving the user experience by following the correct response (Liu, 2018). Through this, steps can be taken to optimize images by using different network foundations and devices, and by using animations and courts to ensure that applications can load in less time.

UX Design Frameworks

According to Kim (2018), A proper process needs to be followed to coordinate the various activities related to the design of the mobile application and thereby ensure that a proper user experience can be created through the use of digital platforms. Part of managing these platforms is ensuring that a designer knows and understands their user data. That's why the designer must test them by generating different types of ideas and photos and take steps to ensure that the user can benefit from them. All these factors help in improving user feedback and can play an important role in executing the correct discovery and problem-solving process by taking a coordinated initiative. By using these focuses, the user can make progress

in knowing and understanding the needs characteristics and behaviors associated with different types of products (Flora, 2014). The numerous design-related futures are increasingly intricate, which has a detrimental impact on the application's usability for users and other features and so lowers user satisfaction. Sometimes a user may be given an application that has an extremely subpar architecture. Consequently, a user may encounter numerous challenges in locating accurate information, which in turn causes him to encounter challenges when executing different parks and other navigation-related tasks. Consequently, the user is unable to navigate the application effectively, which makes it more difficult for him to accomplish his objectives. Again, all of these problems are deemed more difficult for the user if he has any accessibility concerns. Because of this, the application is not correctly calibrated for the user, which causes issues for the user because certain aspects, like the correct brightness, don't operate (Moore, 2018). They encounter difficulties putting different forms of care or other existing cognitive qualities into practice. possibly Finally, a consumer may become somewhat irritated with the program if it operates poorly and takes a long time to load.

According to Hoehle (2015), a Designer requires prior knowledge of how a user will use an application in different environments. Again, as a part of these design activities, a designer can use atomic design methods and through this, by dividing the design into different modules, they can be scaled off and included in the design. On the other hand, by using Lean UX (user interface) design, designers can design a wide range of environments and play an important role in creating and testing various design-related protocols and using them in real life. They help in continuous improvement for a customer and take feedback from different parts of real life.

Problems in existing UX design

According to Dirin (2018), By designing UX a user may face various types of problems in performing all the activities that are undertaken in designing mobile applications. One of these problems is executing the design in a way that does not bring any benefit to the user. By this, a designer fails to identify the issues that the user needs to take seriously, thereby demonstrating a designer's failure to understand the user's needs and behaviour. One of the reasons behind this is that a designer carries out his design activities without proper research and analysis and thereby creates a solution that is unable to bring about any effective action or change. felt that new designs failed to give him full support in implementing them. In

some cases, a designer follows all design patterns that provide different outputs for different screens. All these factors can negatively affect a user's experience and play an important role in confusing the user's actions (Indriana, 2017). So a designer must carry out interface design with proper information and by following proper design guidelines and these can contribute effectively in dealing with all the challenges involved in performing various types of interactions.

From the perspective of Ma (2014), In some cases, a designer follows certain features in application design, through which a large amount of information is displayed to the user within the interface. As a result, the user faces various problems in understanding these data and these contribute to creating a negative experience. By doing so, the various futures related to design become very complex and these significantly affect the user's satisfaction by negatively affecting the usability of the application for a user and the other functions it has. In some cases, an application may be created for the user, which uses a very low-quality architecture. As a result, a user may face multifaceted problems in finding correct information and these make him face problems in correctly executing various parks and other things related to navigation (Punchoojit, 2017). As a result, the user fails to navigate properly within the application and these hinder him in achieving his goals. Again, if the user has any kind of accessibility issues, then all these issues are considered more complicated for him. As a result, the application is not properly calibrated for the user and they create problems for the user due to the correct brightness or other features not working. They face problems in implementing various types of care or other cognitive features that are present. Finally, if the application does not perform effectively and takes a long time to load, it can create a kind of frustration for the customer. Through this, the trustworthiness and reliability of the application can create doubts for the customer and these affect the image of the application significantly. Performance optimization of the application should be taken on occasion and efficiency should be brought into the operation of the application by using different types of image compression or optimization or by using a better server (Nguyen, 2015). Its content plays an important role in improving user experience and increasing engagement by keeping them engaged with the organization.

Theoretical Frameworks

Various theories and frameworks have been used to develop mobile applications and one of them is the cognitive load theory. One of the features of the human brain is that it can process

a limited amount of information. This means that those involved in application design must use nut design and take steps to ensure that the user has a simple interface and that the user can stay focused. For this the user must provide certain information and all these factors help in achieving usability and a fine user experience for the user (Feng, 2019).

Information processing theory can be used as a framework as part of mobile application design, and this theory provides insights into how a person receives and stores information and analyzes that information. Through this, a designer can use all these data and one can get a proper idea about how to collect the data of the application and store it and use it in future operations. All these things can support the designer in building his application importantly. For example, by using this theory, a designer can divide the information he receives into many small and meaningful parts (Moore, 2019). This makes it possible to create a variety of visual cues within the application, and these can significantly aid in processing information and enhancing the user's computer.

Research Gap

Identifying the gaps in the research as part of the research process is considered an important issue. As part of the literature review of this study, important gaps were observed in the data that were found. The observations made as part of the research, there is a lack of practical things in them. As a result, the collected data contained less factual information and thus did not succeed in accurately finding the factual information needed to perform the application design. All these factors act as various obstacles to developing a new mobile application (King, 2016). Therefore, integrated processes have been followed to eliminate all these problems and through this, necessary steps have been taken to bring out the results of the research by using real data.

Conclusion

Mobile applications assist in offering the general public a range of methods to receive guidance and feedback regarding their personal lives, enabling users to monitor their daily physical activity, dietary habits, and other factors. Nonetheless, a lot of people run across different issues when attempting to use fitness and health-related apps correctly. These elements all have a big impact on the user experience but don't give users the incentive they need to interact with the app (Flora, 2014). All of these problems are seen as a form of

impediment to the management of mobile applications, and they pose significant difficulties for the user in terms of practical design problems. Effective customer-focused measures taken by mobile applications can greatly benefit the organization by strengthening the relationship between the application and the user, keeping them interested in the business, and boosting overall productivity. Can This study examine several UX (user interface) design-related concerns on health and fitness apps and how to enhance the user experience by designing an interface that effectively engages the user Considerable advancement in the knowledge and comprehension of those subjects can be achieved by taking action to connect them to applications (Dove, 2017). The aforementioned themes of almonds are reviewed in this literature review, and they greatly advance our knowledge and comprehension of the numerous subjects.

Chapter 3 - Research Methodology

Research Method

Research methods play a very important role in carrying out activities related to research. Due to this, it is possible to get the necessary review of the steps taken by the researcher in carrying out all the activities related to his research. Through this, it is possible to get a plan related to how to implement the research techniques followed by the researcher in carrying out all the issues related to his data collection or data description. It helps in importantly coordinating all these matters. In carrying out the activities of this research, a secondary research process has been followed and data has been collected from various secondary sources (Gray, 2018). Qualitative data analysis process has been used to analyse the data. All these factors help in getting accurate results from the research and they play an important role in integrating the results of the research.

Research Philosophy

Proper research plays a very important role in carrying out various research-related issues. The interpretivism process has been followed to carry out various activities of this research. Here all the literature employed for the research and the data collected by using the secondary method of the research are properly analysed. Steps have been taken to realize the results obtained from all these data. The process of interpreting the results obtained from this realization is followed. Thus, by taking steps to bring results through the correct interpretation of research data, it has become possible to review a large amount of data in good time (Moran, 2018). At the same time, the research method helped the researcher to understand in detail the underlying meanings that were included in all these data. All these things play an important role in getting the correct results from the research (Liu, 2018).

Research Approach

Another important aspect of research is the use of a research approach. Through this, it is possible to determine the right approach for the research and it helps significantly in improving the research process. A deductive approach has been adopted to carry out the activities of this research. Through this, it was not necessary to provide any kind of training in advance to all the data that the researcher has used in the research work and the data that has been analysed. As a result, the researcher is able to directly use the data to find

information (Valentim, 2017). These factors helped in the efficient use of research time and played an important role in integrating research findings.

Data Collection

Data collection is recognized as an acceptable subject for carrying out the activities of various parts of the laboratory. Through this, the researcher collects various types of information for the research work and includes them in the research work. Here the secondary research method has been followed to carry out the various activities of the research and in order to follow this secondary research activities the future has collected data from various types of secondary sources. As a part of all these activities, the researcher followed the process of collecting data from various books, journals, etc. For this, various types of online platforms have been used and information has been collected from there (Weichbroth, 2020). This process of research has helped the researcher significantly and through this, the researcher has been able to collect information from many sources at the time. By using all these subjects, the researcher has been able to review the various existing studies in detail and follow the process of bringing the results from the existing studies by integrating the content of the research. As a result, it has been possible to gain an idea of how the existing studies provide information on various topics (Ma, 2014). These subjects have made progress in carrying out activities and they play an effective role in bringing expertise to the research process.

Inclusion and Exclusion Criteria

As part of the research process, the researcher has used various scholarly databases and collected information from there. One of these databases is Google Scholar and the researcher followed the necessary process to find the information inside this database and took the steps to bring the results. However, some conditions have been imposed in collecting the data here and steps have been taken to find the right type of research paper by reviewing all these conditions. Initially, all the research papers obtained from the selection process were abstractly analysed and steps were taken to find out whether they are relevant to the research topic (Yu, 2016). As a result, only those research papers that were reviewed on the user experience of mobile applications were included. In the next phase, only those papers that described health and fitness related mobile applications were included. On the other hand, all the research papers written in different languages other than English were not included in the

research work and only the English language written questionnaires were included in the research program. On the other hand, those research papers that are more than 10 years old have been excluded from the research review process. As a result, no research paper was included in the study before 2014 and review process was not performed on the research papers published before 2014. In this way, the right activities have been taken in finding the right research paper to implement the various issues related to the research (Nguyen, 2015). By following all these processes, it is possible to play an effective role in bringing about the right results.

Search Strategy

In relation to research, using a search strategy is considered effective in order to properly implement strategies. The researcher must follow the right type of search process to find the data collection. A variety of keywords have been used for this purpose and these keywords have been followed in such a way that they can help in finding information related to research trophies. On the other hand, to use all these keywords effectively, different types of operators are used which are known as Boolean operations (Kushendriawan, 2021). Researcher and operator are used to carry out the research activities and thereby perform searches to come up with integrated results by using the two keyworks together. The researcher and operator used to follow the same process and thereby take steps to find results from one of the two keywords (Feng, 2019). These factors have helped in finding the right results and by following such a search process it has been possible to make significant progress in finding results relevant to the subject of the research.

Data Analysis

Another important step in carrying out research-related activities is to analyse the data. The researcher followed the qualitative research process to analyse the necessary data along with the research. Through this, the researcher has analysed the data obtained from various secondary sources and included them in the research work. In order to do all this correctly, the index of research papers related to them has been analysed and the research papers that can fulfil the necessary conditions have been included in the research program (Kascak, 2014). For this reason, multiple research papers were reviewed by following a detailed strategy and those research papers that were found to be repetitive were excluded from the review process. Thus, a certain number of research papers were included in the review

process and the data analysis process was performed on them. A coding process was initially followed to analyse all these data. For this reason, steps have been taken to create the right type of codes from the research papers and later by using these codes, different types of themes have been created from there. Later, by using these themes, the studies were reviewed and from there it was possible to take action to generate results (Kalimullah, 2017). By following all these points, it is possible to implement the process related to data analysis and they have significantly assisted the researcher in coming out with correct results from the research.

Ethical Consideration

Ethical issues play a very important role in carrying out the activities of various parts of the research. Here, in carrying out the various activities of the research, the researcher has given importance to the ethical policy and therefore the researcher has analysed the reliability and reliability of the collected data. Through this, all the sources that have been used to collect the data, have been verified by following different types of cross-referencing processes and steps have been taken to confirm the selection of research sources (Vermeeren, 2016). At the same time, as a part of implementing the ethical policy, the researcher did not provide any information in his research work that could damage the image of any organization or work revealing any kind of personal information. On the other hand, as a part of deep ethical activities, the data collected through research has been safely stored. As part of the secure data storage process, the researcher used cloud servers and followed a password protection system to store the data by using Google Drive. After that, the information can be taken into the hands of a second party or this information cannot be used for any commercial purpose, all the issues have been confirmed (Lestari, 2014). By implementing all these issues, the researcher has taken steps to ensure the ethical issues of his research activities and this help significantly in coordinating the research activities.

Chapter 4: Design and Evaluation

The prototype has been designed using Figma. The below link shows the Figma design of the mobile application:

https://www.figma.com/file/hBrle3yhQhoqVag2KFlBWT/Fitness-App?type=design&node-id=0-1&mode=design

Prototype

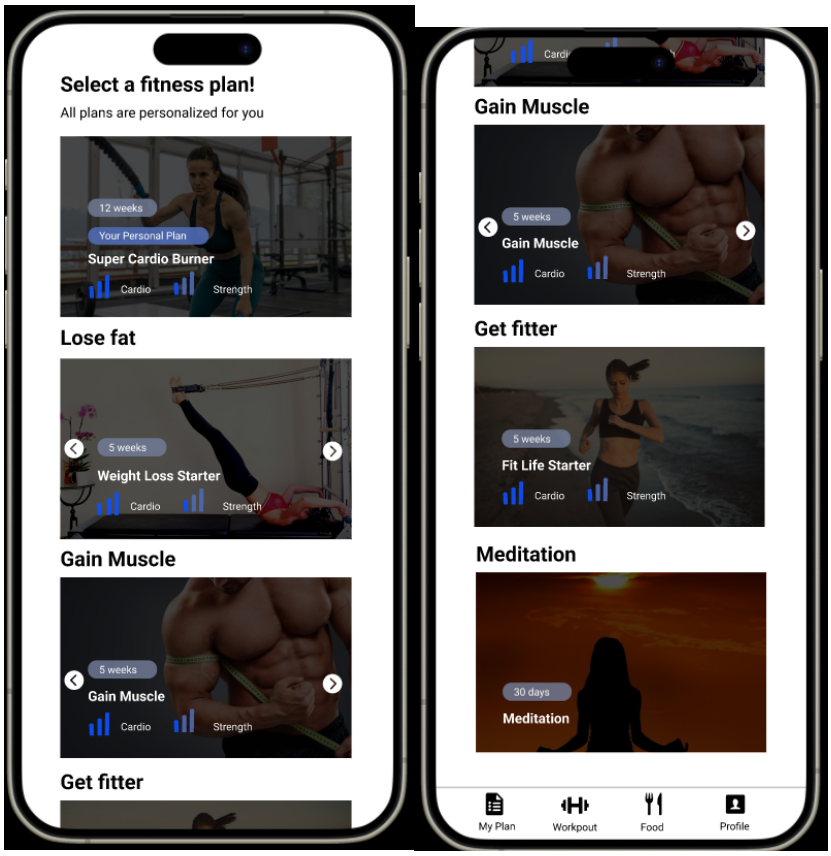


Figure : Home screen

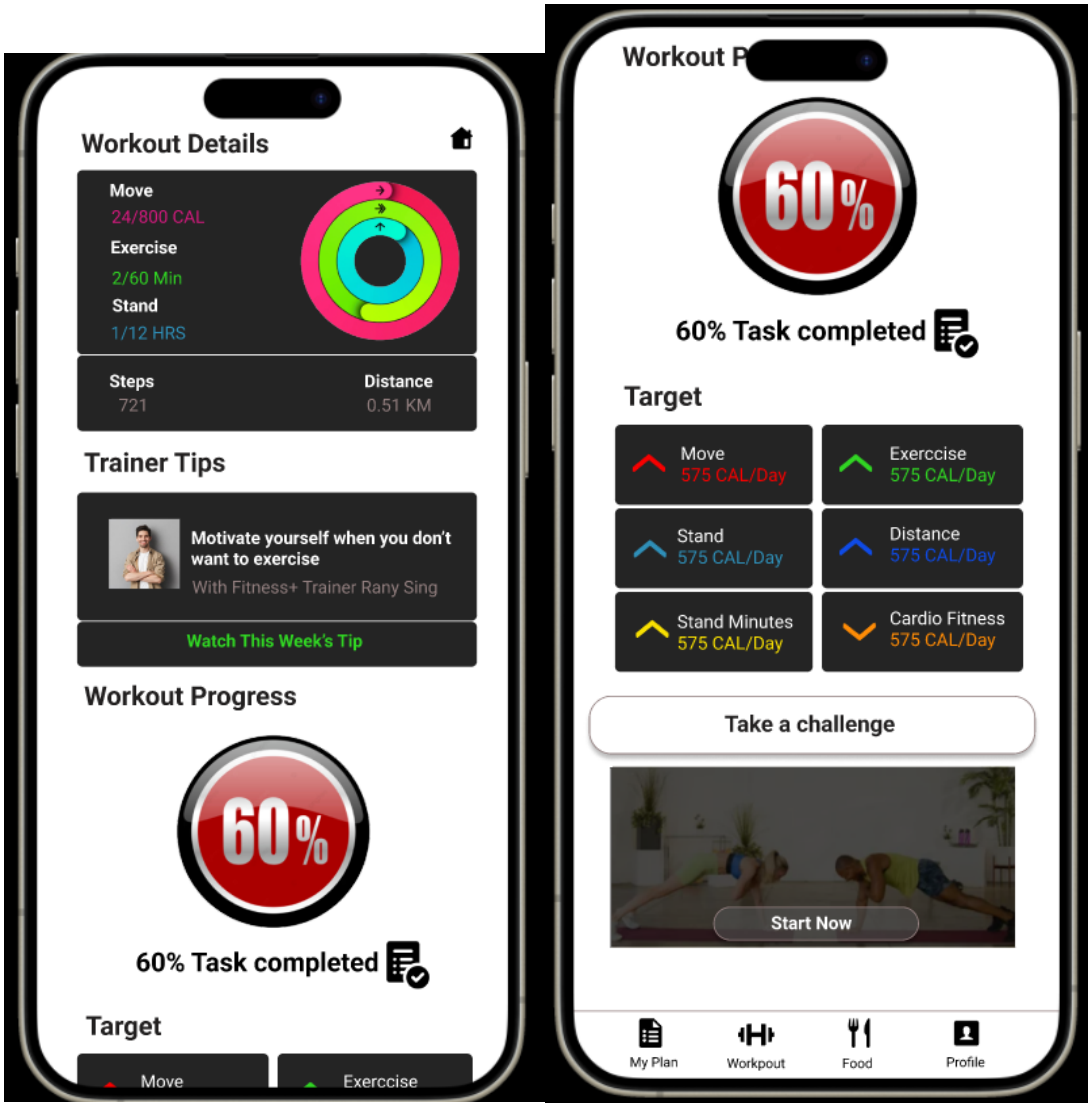


Figure: Workout details screen

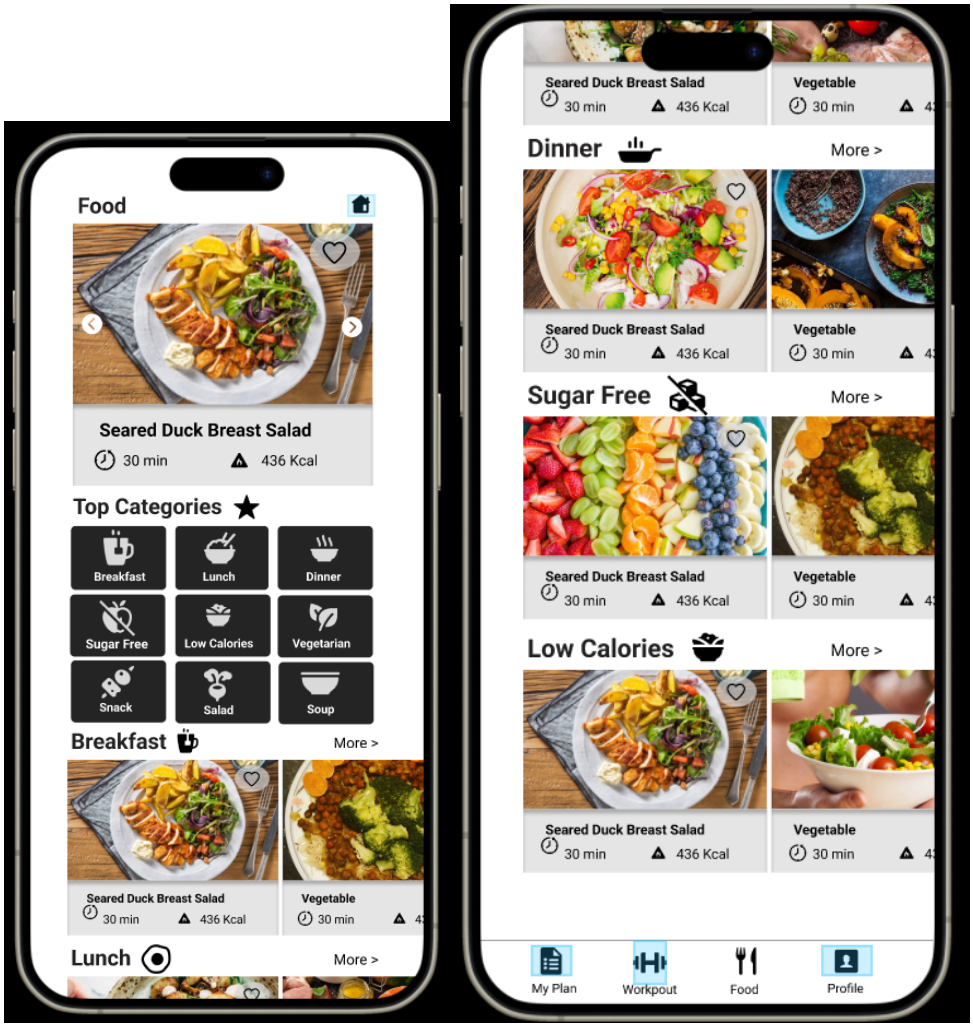


Figure: Food screen

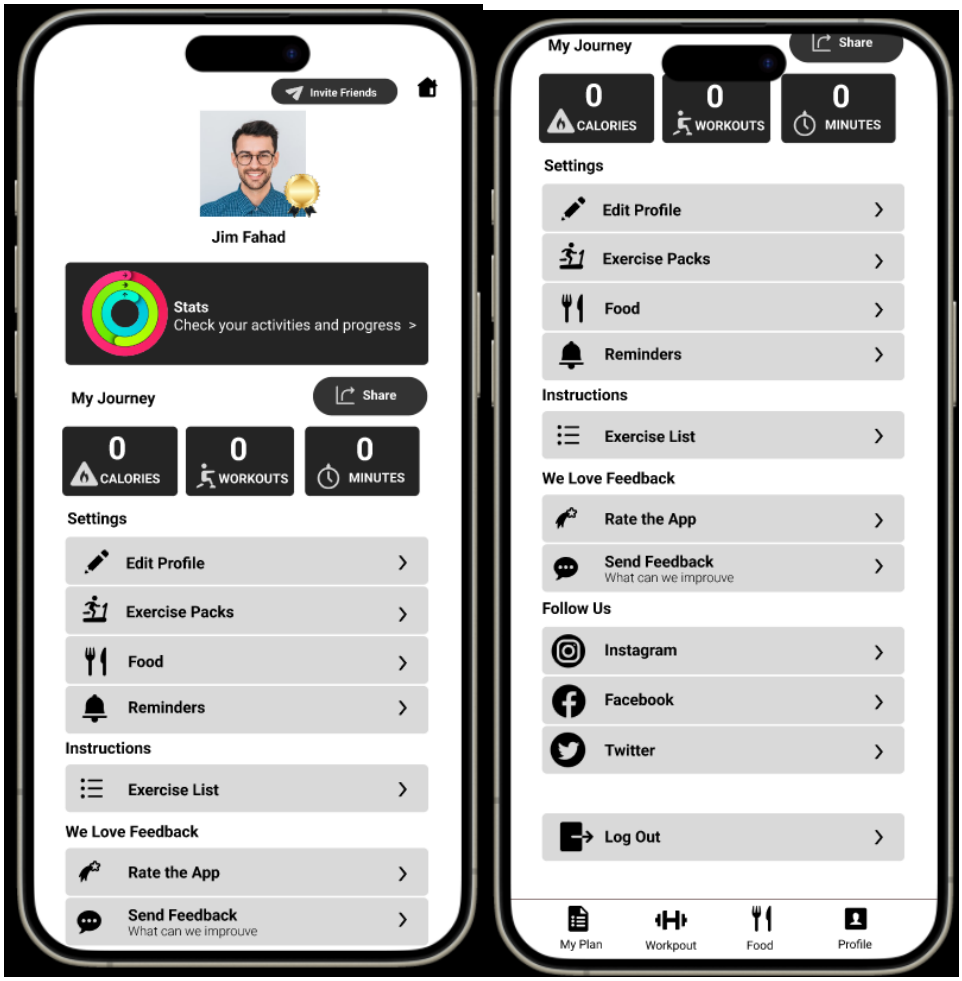


Figure: Profile screen

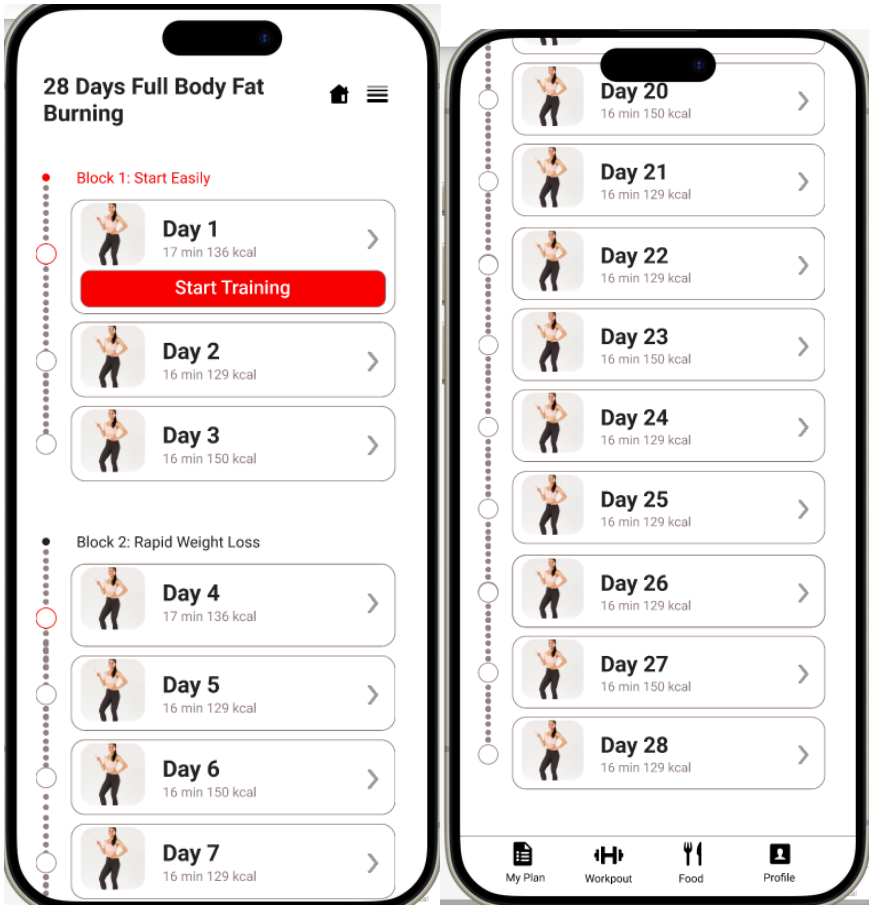


Figure: Video tutorial screen

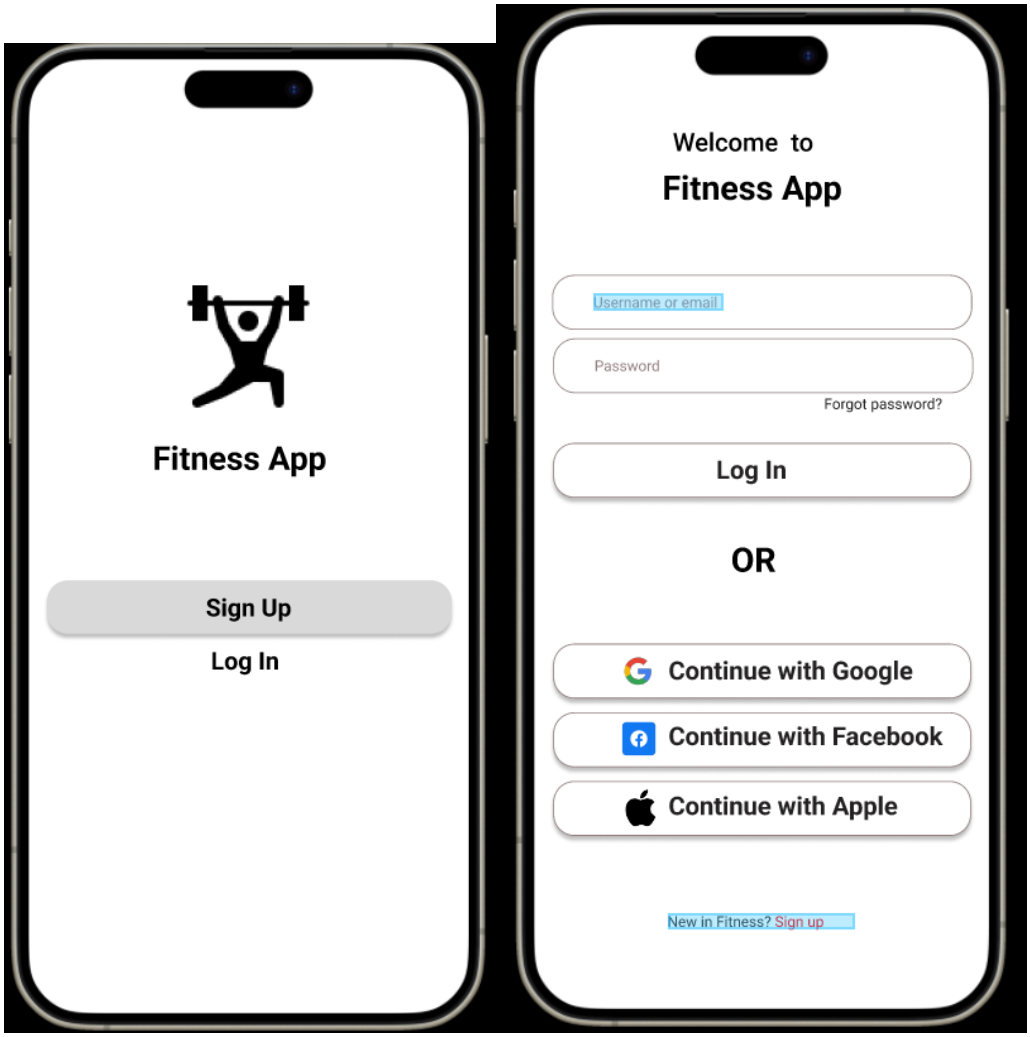


Figure: Login screen

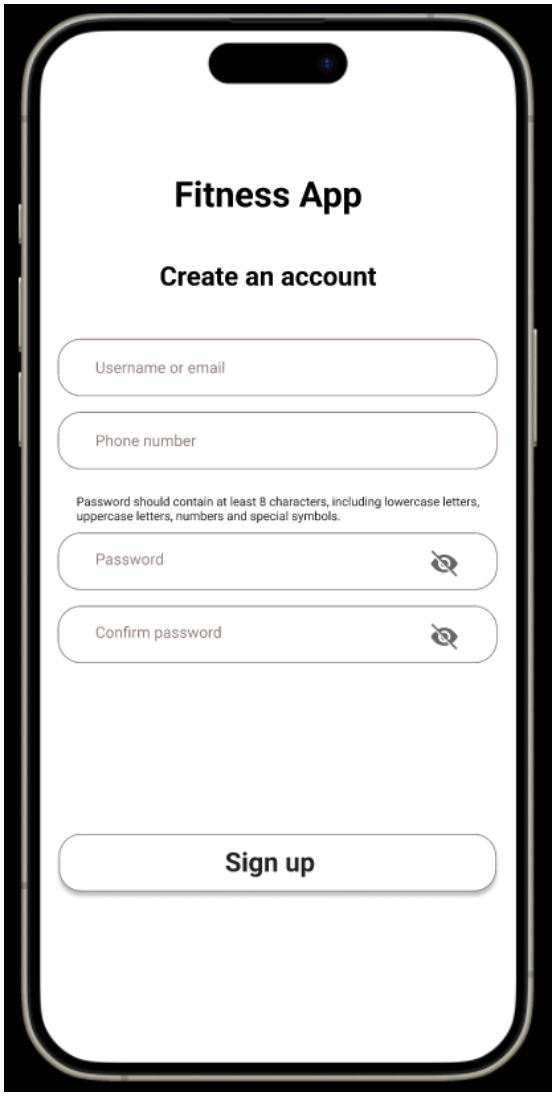
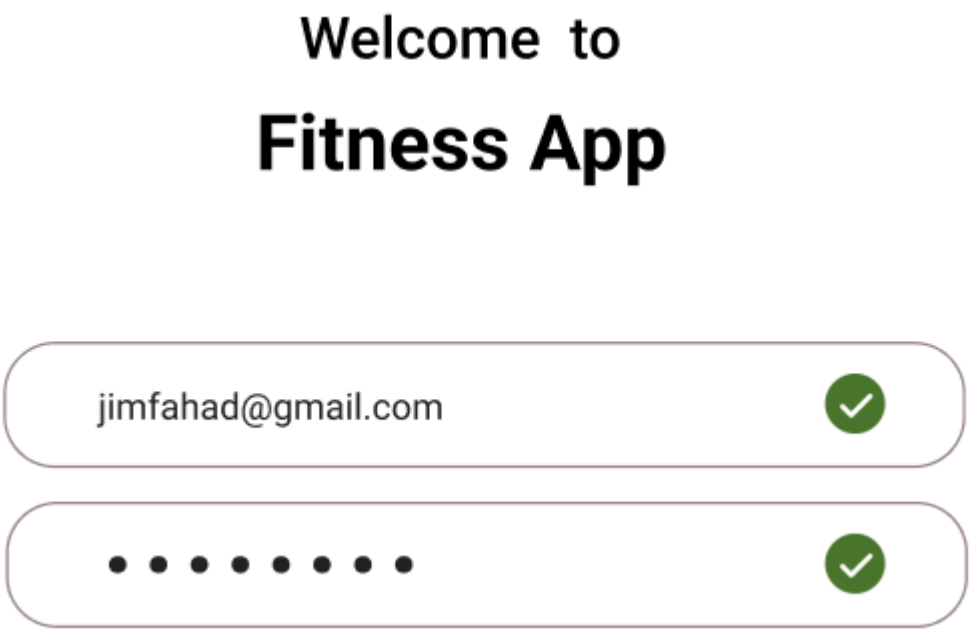


Figure: Account screen

Evaluation

1) Visibility of System Status

The designed system informs the users what's going on in the application and appropriate feedback is given. For example, in the app, if user enters email and it's valid then a tick mark with green background is shown.



2) Match between the system and the real world

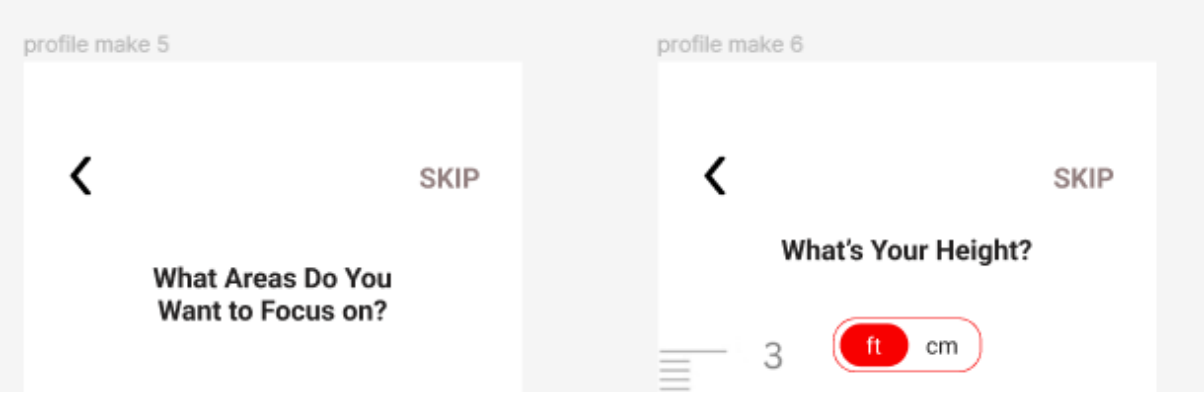
The designed app uses real world phrases, words, concepts which are familiar to the users.



The above icons have been used in the app for navigation and the app shows a notepad type icon for my plan, a workout equipment for workout screen, cutlery for food screen and person abstract image for profile screen.

3) User control and freedom

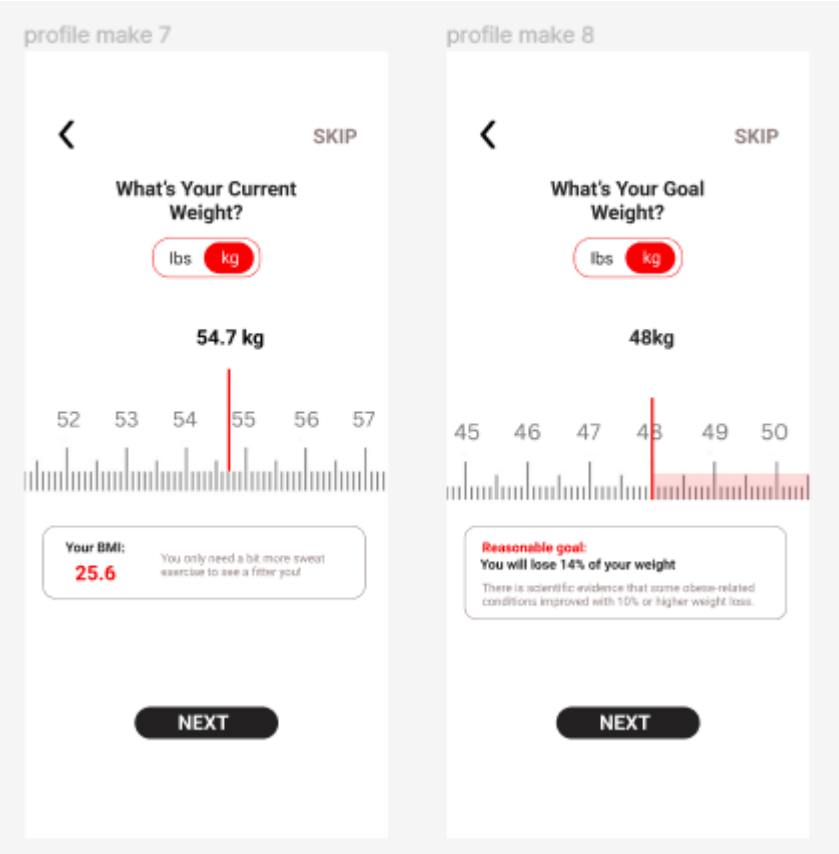
Sometimes users perform actions that are done by mistake also they need some options like cancel in a form so that they can leave the unwanted action.



In the profile screens, the users have the option to go back to the previous page to make any necessary changes required. They can also skip a screen like adding height, main goal, etc.

4) Consistency and Standards

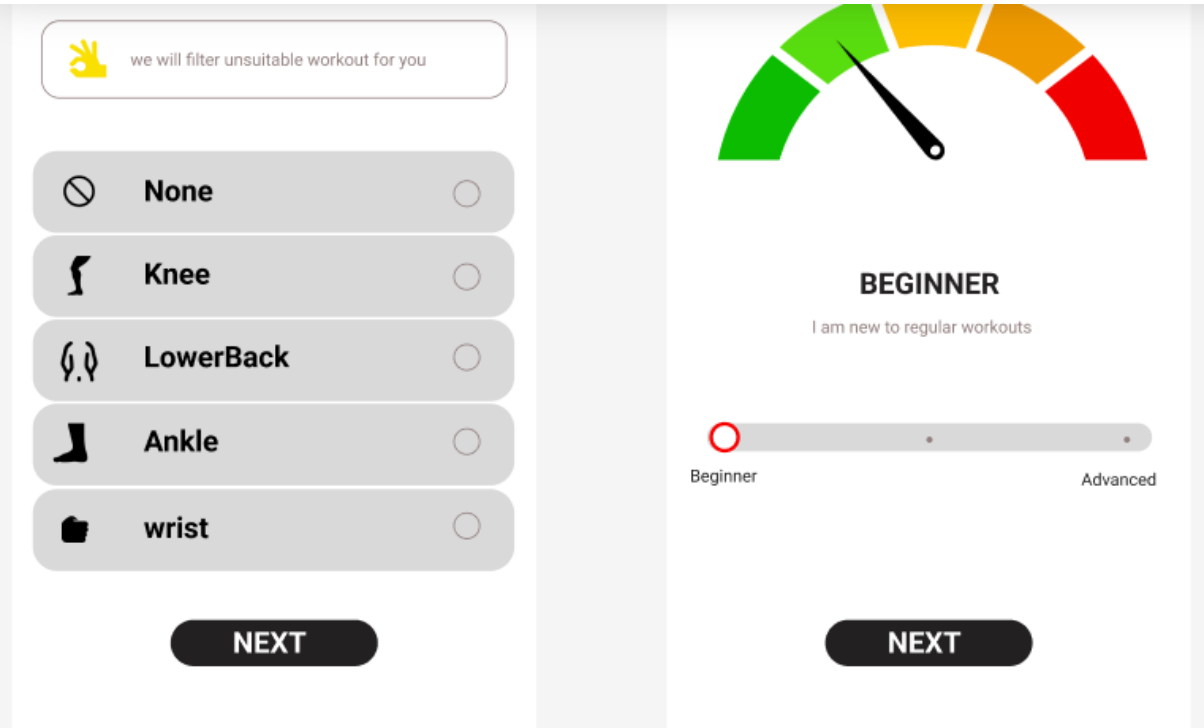
In an app, the design should have consistency among various screens so that the users don't get confused and faces problems in using that app.



In the app, consistency has been maintained like adding left arrow for back button in the top left corner and adding SKIP word for skipping any screen. Also, next button for adding data and going to the next option.

5) Error prevention

Showing error message is a good approach but trying best to avoiding error is the best approach. The errors in mobile apps are eliminated through error-prone conditions, or checks and presenting users with a confirmation option before they complete an action.



Instead of adding text box for adding options appropriate checkboxes and radio buttons have been added so that the user doesn't mistakenly add something else.

6) Recognition rather than recall

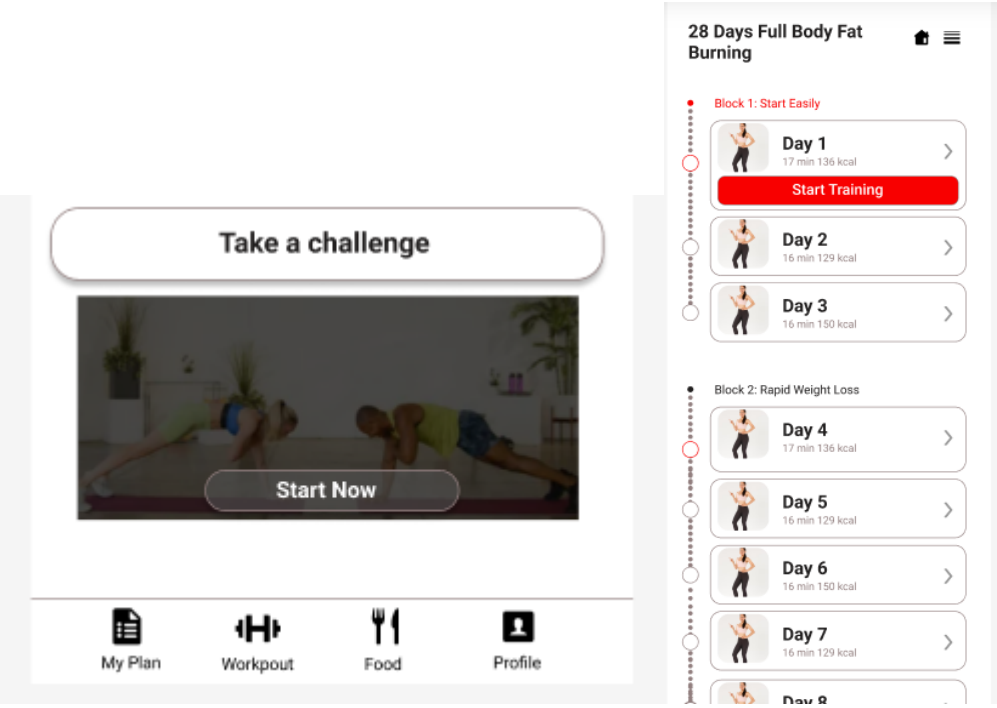
The mobile app users want convenience and they want to recognise rather than recall. It's useful for people if the elements, options and actions are visible.



The above screenshot shows that navigation bar has been used for this purpose and the users don't have to recall and they just can recognise.

7) Flexibility and efficiency of use

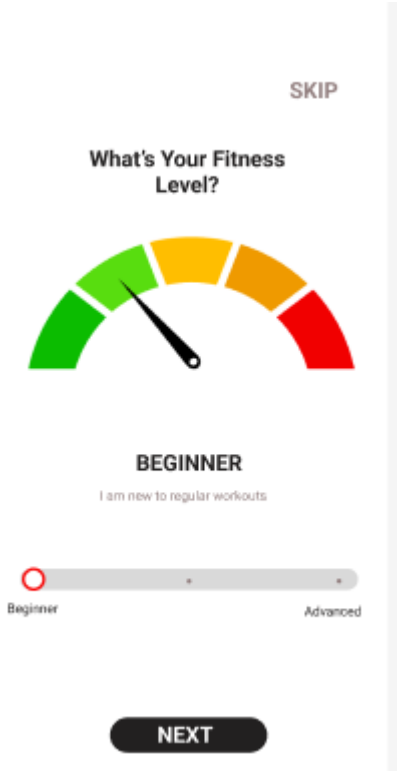
The users of the mobile app can easily use a system if the system provides shortcut for experienced users so that the users can easily interact with the system.



The users of the designed mobile app have such shortcuts like take a challenge and can skip tutorial days and move from day 1 to day 11.

8) Aesthetics and minimalist design

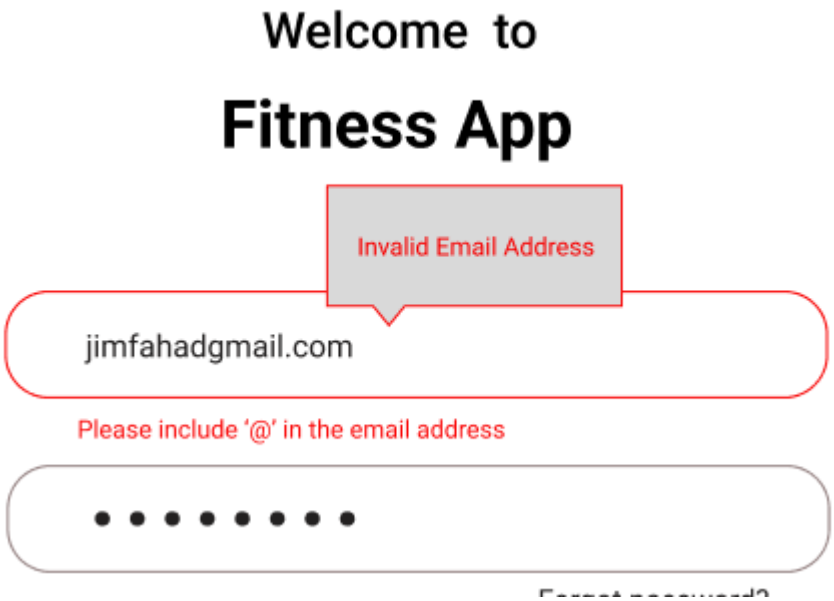
Though aesthetic is a important part of design however, the design should be minimal in mobile app and no irrelevant information should be avoided.



For example, the mobile app screen shows that it only contains graphics and radio buttons instead of adding too much information on this mobile screen.

9) Help users recognise, diagnose, and recover from errors

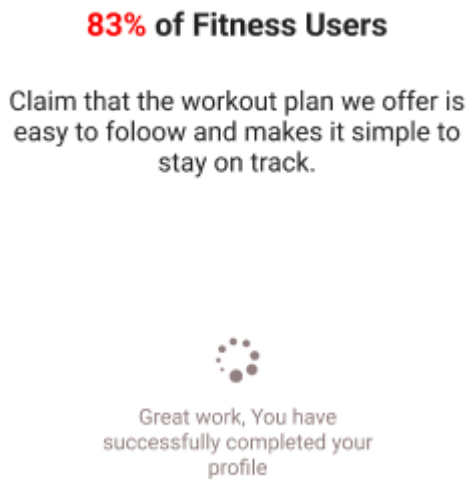
In the applications, the error message should be shown in plain text and the app should precisely show the problem and also suggest a solution.



In the mobile app design, if the user doesn't provide valid email address, then the users will see error message helping them recover from the error.

10) Help and documentation

Though it's best to make a system which doesn't need any additional explanation, it's a good practice to add proper documentation so that the users can easily understand and use the application.



After successfully completing the profile, the users will get a proper documentation that the user can now claim the workout plan and stay on track.

Chapter 5: Conclusion

The development of a mobile health and fitness application project is aimed to enhance user engagement and motivation by using a user-centric approach that focus on UX design principles. The common issues in the health and fitness app scenario include, poor customer engagement and usability related challenges. This project aimed to create a comprehensive solution that benefits users to effectively pursue their health and fitness goals. By seamless integration of various health-related functionalities and media, the application functions to provide users with the tools to track daily activities, plan meals, and monitor progress.

Additionally, the project aimed to expand the gamification and motivational features to enhance customer engagement and promote long-term app usage. The research project was completed by firstly reviewing existing literatures and existing health and fitness mobile apps. Through enough understanding of UX design principles and existing mobile app UX issues a prototype was designed using Figma. While designing the system, Nielsen's 10 heuristics and WCAG principles were followed. This was the result of the research of last few months and I was able to design a better mobile app UX design following the UX guidelines. The designed system has been evaluated based on Nielsen's 10 heuristics and WCAG. The evaluation shows the design follow the usability heuristics and WCAG principles.

Finally, the mobile health and fitness application project aims to address the multifaceted challenges encountered by users within the use of health and fitness apps. By focusing its approach around user experience (UX) design principles, the project search to create a platform that not only aligned user interactions but also cultivates a supportive and motivational environment for individuals to seek their wellness and fitness goals. Overall, by prioritizing user experience and employing innovative approaches, the mobile health and fitness application project drives to revolutionize the way that individuals engage with their health and ultimately contributing to improved well-being and lifestyle of the user and the application seeks to benefits the users to maintain a sustainable healthy lifestyle.

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